

- 19** An optical telescope just resolves two distant yellow stars. They are subtended at an angle of 2.0×10^{-6} radians at the telescope. The wavelength of yellow light is about 590 nm.

Which statement is correct?

- A** The aperture of the telescope has a diameter of about 0.30 m.
- B** The telescope can be used to distinguish two red stars at similar distance.
- C** The resolving power of the telescope is independent of the wavelength of light.
- D** The telescope can be used to distinguish stars subtended at an angle of 3.0×10^{-7} radians.

- 20** A 5.0 C point charge is initially stationary at a point P in a uniform electric field of